

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 June 2024

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2024 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. In a class of Gifford High, $n(A) = 15$ (play soccer), $n(A \cap B) = 2$ (play both). How many play soccer only? **[2]**

2. Work out $3/7 \div 2/2$. **[3]**

3. Share \$33 in the ratio 4 : 7. **[3]**

4. Find the HCF and LCM of 36 and 42. **[3]**

5. Expand and simplify $5(x + 1) - 5x$. **[3]**

6. Simplify $2^5 \times 2^3$. Leave as a power of 2. **[3]**

7. Work out $(5 \times 10^6) \div 10$. Give the answer in standard form. **[3]**

8. Overnight in Harare the temperature was 4°C . By 14:00 it was 22°C . Find the rise. **[3]**

9. Express 90 as a product of its prime factors. **[4]**

- 10.** Find 40% of 250. **[3]**
- 11.** Find the midpoint of (6, 1) and (9, 5). **[3]**
- 12.** Factorise $x^2 - 9$. **[4]**
- 13.** Work out $(2 \ 2 ; 3 \ 2)(5 ; 3)$. **[4]**
- 14.** Make h the subject of $A = 2\pi r(r + h)$. **[4]**
- 15.** A bag contains 3 red, 2 blue and 4 green beads. $P(\text{not green})$? **[3]**
- 16.** Find the mean of 12, 11, 8, 4, 6. **[3]**
- 17.** Area of a circle radius 14 cm. Take $\pi = 22/7$. **[4]**
- 18.** A cuboid water tank measures 12 m by 3 m by 6 m. Find its volume. **[3]**
- 19.** A regular polygon has 9 sides. Find each exterior angle. **[3]**
- 20.** Solve $4x + 5 = 37$. **[3]**
- 21.** In triangle ABC, $\hat{A} = 66^\circ$ and $\hat{B} = 36^\circ$. Find $\hat{C}$. **[3]**

- 22.** A sequence is 9, 11, 13, 15, ... Find an expression for the n th term. **[4]**
- 23.** A right-angled triangle has shorter sides 16 cm and 30 cm. Find the hypotenuse. **[4]**
- 24.** Factorise $x^2 + 7x + 10$. **[4]**
- 25.** In right-angled triangle PQR, $\hat{P} = 30^\circ$ and hypotenuse $PR = 8$ cm. Find PQ, opposite 30° . **[4]**
- 26.** A trader at Mbare Musika buys a box of tomatoes for \$25 and sells for \$45. Percentage profit? **[4]**
- 27.** Solve $x + y = 4$ and $x - y = 0$. **[4]**
- 28.** Write the equation of the line with gradient 2 passing through (0, -1). **[3]**
- 29.** A triangular plot has base 12 m and perpendicular height 7 m. Find its area. **[3]**
- 30.** Solve $3x + 5 < 23$. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).